Spitfire® Pro XP Ultrafast Amplifier

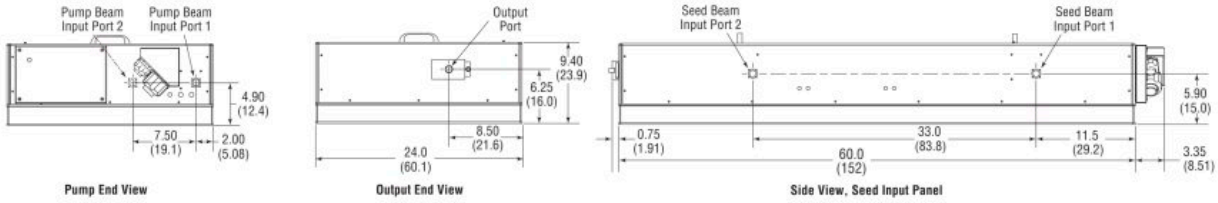
Spitfire Pro XP 4W (1W) Specifications[1]

Output Characteristics	Spitfire Pro 35F	Spitfire Pro F	Spitfire Pro PM
Pulse Width[2, 4, 5]	<35 fs	<120 fs	<2 ps
Repetition Rate[3]		1 kHz, 5 kHz or 10 kHz	
Pulse Energy[2] (1 kHz)		>4.0 mJ (>1 mJ)	
Pulse Energy[2] (5 kHz)		>0.8 mJ (>0.2 mJ)	
Pulse Energy[2] (10 kHz)		>0.4 mJ (>0.1 mJ)	
Pre-Pulse Contrast Ratio[6]		>1000:1	
Post-Pulse Contrast Ratio[7]		>100:1	
Energy Stability[8]		<0.75% rms over 8 hours	
Tunability[9]	780–820 nm	750–840 nm	750–840 nm
Transform Limit[10]		<1.5 x transform limit	
Spatial Mode		TEM_{00} (M^2 <1.3 on both axes)	
Beam Diameter (1/e2)		7 mm (nominal)	
Polarization		Linear Horizontal	

Notes

1. Due to our continuous product improvements, specifications are subject to change without notice. The specifications only apply when Spitfire Pro is operated with recommended Spectra-Physics seed and pump lasers.
2. Applies at peak wavelength (800 nm).
3. The desired optimum repetition rate should be specified at time of purchase. If optimum performance is required at more than one repetition rate, additional optic sets may be required. Any system can be operated (with the same energy per pulse) at reduced repetition rates through internal divide-down electronics.
4. A Gaussian pulse shape (0.7 deconvolution factor) is used to determine pulse width (FWHM) from autocorrelation signal as measured with Newport PulseScout® autocorrelator.
5. For other pulse widths, please contact Spectra-Physics.
6. Defined as the ratio between peak intensity of output pulse to peak intensity of any pre-pulse that occurs >1 ns before the output pulse. For higher contrast ratio, please contact Spectra-Physics.
7. Defined as the ratio between peak intensity of output pulse to peak intensity of any pre-pulse that occurs >1 ns after the output pulse. For higher contrast ratio, please contact Spectra-Physics.
8. Applies at peak wavelength: % rms with near Gaussian distribution.
9. For wavelength extension through SHG, THG, FHG or OPA, please contact Spectra-Physics.
10. Assuming Gaussian pulse shape.

Spitfire Pro XP Dimensions



Dimensions in inch (cm)

Spectra-Physics® Lasers Sales: 1-800-775-5273 sales@spectra-physics.com www.newport.com/lasers

Newport Experience | Solutions
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Newport Corporation, Irvine, California; Mountain View, California; Tucson, Arizona; Evry and Beaune-La-Rolande, France have all been certified compliant with ISO 9001 by the British Standards Institution or TUV.

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